

Government Services ETL Platform — Build Roadmap

Personal working roadmap. Track progress by checking off each step as you go.

Golden rule

Build a **minimal end-to-end pipeline first (MVP)**, then layer enhancements on top. Reach a working pipeline (through Phase 4B) as fast as possible — don't perfect one layer before moving on. If time runs short, you still have a complete, demoable system.

Tech stack at a glance

Layer	Tool
Data sourcing	Kaggle dataset · web scraping (BeautifulSoup / Scrapy / Selenium / Playwright) · Faker
Ingestion (batch)	Azure Data Factory
Ingestion (streaming)	Azure Event Hubs
Data lake	Azure Data Lake Storage Gen2 (Bronze / Silver / Gold)
Transformation	ADF Mapping Data Flows (Spark-based)
Warehouse	Azure SQL Database (star schema)
Orchestration	Apache Airflow
Monitoring	Airflow + Azure Monitor
Data quality	Great Expectations
Dashboard	Power BI
ML / NLP (stretch)	Azure ML · Azure OpenAI · AraBERT
Version control	Git / GitHub

Phase 1 — Planning & Approval

- ☒ Write project proposal — **Word**
- ☒ Get supervisor approval

Output: approved proposal

Phase 2 — Requirements Gathering

- ☐ Identify stakeholders and use cases — **Word / Markdown**
- ☐ Define functional & non-functional requirements

Output: requirements specification

Phase 3 — Analysis & Design

- ☐ Use Case diagram + system architecture diagram — **draw.io / Visio**
- ☐ ERD and detailed star-schema design
- ☐ Data Flow Diagram (DFD)
- ☐ Dashboard wireframes

Output: design document + diagrams — the backbone before any code

Phase 4 — Implementation

4A · Setup

- ☐ Create Azure Resource Group + budget alert — **Azure Portal + Cost Management**
- ☐ Create Git repository — **Git / GitHub**
- ☐ Acquire data (see *Data sources* below) — **Python · scraping · Kaggle**

4B · MVP — end-to-end pipeline

- ☐ Create data lake with Bronze / Silver / Gold zones — **ADLS Gen2**
- ☐ Land raw data into Bronze — **Azure Data Factory (Copy Activity)**
- ☐ Build warehouse + star schema (fact + dimension tables) — **Azure SQL Database**
- ☐ Clean & standardise (Bronze → Silver) — **ADF Mapping Data Flows**
- ☐ Aggregate & model (Silver → Gold) — **ADF Mapping Data Flows**
- ☐ Load Gold into the warehouse — **ADF (Data Flow sink)**
- ☐ Connect the dashboard + build first KPI — **Power BI**

=== MVP CHECKPOINT === At this point you have a working pipeline from source to dashboard. Everything below is enhancement.

4C · Automation & Quality

- ☐ Orchestrate with DAGs that trigger ADF pipelines (REST API), with dependencies & retries — **Apache Airflow**
- ☐ Add monitoring and failure alerts — **Airflow + Azure Monitor**
- ☐ Add data-quality checks before load — **Great Expectations**
- ☐ Add real-time complaint streaming — **Azure Event Hubs**

4D · Advanced (stretch — your differentiators)

- ☐ Anomaly detection for abnormal complaint spikes — **Azure Machine Learning**
- ☐ Conversational layer (text-to-SQL, Arabic) — **Azure OpenAI Service**
- ☐ Arabic complaint-text analysis (classification + sentiment) — **AraBERT / Python**

Phase 5 — Testing & QA

- ☐ Test the full pipeline and validate data — **SQL validation queries**
- ☐ Document test cases and bugs

Phase 6 — Final Presentation & Reports

- ☐ Finalise documentation + user manual — **Word**
 - ☐ Build the presentation deck — **PowerPoint**
 - ☐ Record a demo video — **screen recorder**
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Data sources (updated approach)

Use **real data** instead of purely synthetic — it is more credible and showcases real skills. The best approach is a mix:

1. **Primary — real complaints dataset (Kaggle / open data):** e.g. **311 service requests** (city open data) or **CFPB consumer complaints**. These map cleanly onto the star schema: agency → ministry, borough/state → governorate, complaint type → service, plus status and resolution time.
2. **Secondary — web scraping (your specialty):** scrape public ratings/reviews of government services — best angle: **Arabic Google Maps reviews of Egyptian government offices**, or a public forum. This makes the "multiple sources" claim real, showcases your scraping skill, and feeds the Arabic-NLP stretch goal. — **BeautifulSoup / Scrapy / Selenium / Playwright**
3. **Optional — API source:** pull from an open-data portal API to cover a third source type. — **Python requests**
4. **Synthetic — fill gaps only:** generate any missing dimension/reference data (e.g. a clean governorate table). — **Faker**

Timebox the data hunt. The pipeline is the deliverable, not the dataset. Pick a solid source quickly and move on to building. Scraped/Kaggle files land in the data lake (Bronze), then ADF picks them up — so this fits the Azure flow cleanly.